



PROGRAM SETUP

Choose a galaxy or stars backdrop and delete the default cat



STEPS TO FOLLOW

1. Open scratch.mit.edu and click "Create"
2. Right-click the cat sprite → Delete
3. Click "Choose a Backdrop" (bottom-right)
4. Search "Stars" or "Galaxy" → click the dark starfield
5. Sign in to auto-save your project!

TIP

A dark background makes the yellow stars pop visually. Contrast is a key principle in game design!

DID YOU KNOW?

The Scratch stage is 480×360 pixels.

$x=0$, $y=0$ is the dead centre.



OBJECT PROPERTIES

Add a Bowl sprite, set size to 80, and position it near the bottom



SCRATCH BLOCKS — ON THE BOWL SPRITE

when  clicked

go to x: (0) y: (-150)

set size to (80) %

TIP

y: -150 places the bowl near the bottom. The stage goes from y=-180 (bottom) to y=180 (top).

DID YOU KNOW?

Every object in a game starts with initialization code that sets position, size, and state before the player can interact.



EVENT HANDLING · LOOPS

Left/right arrow keys slide the bowl across the screen; edge stops it



SCRATCH BLOCKS — ON THE BOWL SPRITE

when  clicked

go to x: (0) y: (-150)

forever

if <key [left arrow] pressed?> then

change x by (-10)

if <key [right arrow] pressed?> then

change x by (10)

if on edge, bounce

TIP

Set Bowl's rotation style to "left-right"

in Sprite Info so "if on edge, bounce"

never flips it upside-down.

DID YOU KNOW?

This input-polling pattern inside a forever loop is the game loop — it runs in Unity, Godot, Pygame at 30–60 fps.



OBJECT TEMPLATES · HIDING

Add a Star sprite and hide it on flag click — it will be the clone template



SCRATCH BLOCKS — ON THE STAR SPRITE

STAR: *when*  *clicked*

hide

TIP

Any sprite that uses clones **MUST** hide its original first. Otherwise the "master" copy sits frozen on the stage forever.

DID YOU KNOW?

This template-hide pattern maps directly to a class definition in OOP — the class sits in memory invisibly until you instantiate (clone) it.



CLONE INSTANTIATION

Make the master Star create clones that fall from top to bottom
and delete themselves



SCRATCH BLOCKS — ON THE STAR SPRITE

STAR master: when  clicked

hide

forever

wait (1) seconds

create clone of [myself]

STAR clone: when I start as a clone

show

repeat until <(y position) < (-170)>

change y by (-5)

delete this clone

TIP

"delete this clone" is essential —

without it, thousands of invisible off-screen stars pile up and slow the game to a crawl.

DID YOU KNOW?

"delete this clone" is the game-object lifecycle: create → update → destroy.

Every bullet, enemy, and particle follows this same lifecycle.



RANDOM NUMBERS

Make each clone appear at a random x position so stars fall from different spots



SCRATCH BLOCKS — ON THE STAR SPRITE

STAR clone: when I start as a clone

go to x: (pick random (-200) to (200)) y: (175)

show

repeat until <(y position) < (-170)>

change y by (-5)

delete this clone

TIP

Try pick random (-170) to (170) so stars never spawn partially off-screen.

DID YOU KNOW?

Random number generation drives loot tables, enemy spawns, procedural worlds, and weather systems. Randomness makes games feel alive.



COLLISION DETECTION • POLLING

Detect when a falling star touches the Bowl — celebrate with a speech bubble



SCRATCH BLOCKS — ON THE STAR SPRITE

(inside repeat-until loop)

change y by (-5)

if <touching [Bowl]?> then

say [★ Caught!] for (0.5) secs

delete this clone

💡 TIP

"touching [Bowl]?" polls every frame

— asking "do these two sprites

overlap?" up to 30 times per second.

🧠 DID YOU KNOW?

Collision detection is one of the most important algorithms in games. Unity calls these "colliders", Godot calls them "collision shapes".



VARIABLES • STATE

Create a Score variable and add 1 every time a star is caught



SCRATCH BLOCKS — ON THE STAR SPRITE

(inside if <touching [Bowl]?>)

change [Score] by (1)

delete this clone

TIP

Right-click the Score display on the stage and choose "Large readout" for a big satisfying counter!

DID YOU KNOW?

A variable is the most fundamental concept in all programming. Python, JavaScript, Swift — every language uses them.



GAME LOOP · LIFECYCLE

Add Lives — lose one when a star escapes; game ends when
Lives reach zero



SCRATCH BLOCKS — ON THE STAR SPRITE

(after the repeat-until - star escaped)

```
change [Lives] by (-1)
```

```
if <(Lives) = (0)> then
```

```
  say [Game Over! 🦋] for (2) secs
```

```
  stop [all]
```

```
delete this clone
```

💡 TIP

Show ♥ hearts for lives by right-clicking the Lives display — or display it as "large readout" for impact!

🧠 DID YOU KNOW?

PLAYING → GAME_OVER is a state machine. Traffic lights, login flows, vending machines, and game modes are all state machines.